

Real-Time Approximation of Hydraulic Erosion: A Parameter-Conditioned Deep Learning Approach

Zea Fernandez, J.S.¹

¹Faculty of Basic Sciences, Universidad de Medellín, Colombia

Abstract

Realistic terrain generation relies on hydraulic erosion to form plausible drainage networks and valleys. Lagrangian particle-based simulations offer high-fidelity approximations of sediment transport, yet they suffer from prohibitive computational costs that scale linearly with simulation time ($O(N)$). This dependency renders physics-based erosion unusable for real-time applications. We address this bottleneck with a differentiable surrogate model based on a Conditional Generative Adversarial Network (cGAN). Unlike standard style-transfer approaches, we introduce a parameter-conditioning mechanism: scalar physical constants—such as inertia, capacity, and erosion rate—are spatially broadcasted and stacked with the input heightfield. This forces the network to learn the specific non-linear operators of the physical simulation. Empirical benchmarks demonstrate a $193\times$ speedup for single-instance inference compared to a Numba-optimized physics engine. Additionally, recursive inference tests show that the model maintains temporal stability, simulating up to 10^6 droplets with linear error growth. This framework decouples simulation fidelity from computational latency, enabling instant, physics-aware terrain editing.

Keywords: Procedural Content Generation, Hydraulic Erosion, Deep Learning, Neural Surrogate Modeling, Real-Time Rendering

1 Introduction

Procedural terrain generation faces a longstanding trade-off between physical realism and computational efficiency. Early stochastic methods, such as fractional Brownian motion (fBm) [1, 2], are computationally inexpensive but fail to capture geomorphological coherence. Real landscapes are defined by the history of fluvial and thermal weathering processes that carve drainage networks and deposit sediment over geological timescales—features that random noise cannot reproduce.

Hydraulic erosion simulations remain the standard for addressing this lack of realism. Following the foundational work of Musgrave et al. [3] and Chiba et al. [4], modern approaches largely favor Lagrangian particle-based hydrodynamics [5, 6]. Recent extensions have applied these principles to unstructured triangular meshes to support complex 3D features [7]. In

these models, discrete water droplets traverse the terrain, eroding and depositing material to form dendritic structures. Despite their visual superiority, Lagrangian simulations face a severe computational bottleneck: complexity scales with the number of particles (N) and path length (T). High-resolution results necessitate millions of iterations, introducing latencies that prohibit their use in interactive authoring tools or real-time rendering pipelines.

Data-driven generative models offer a potential alternative. Generative Adversarial Networks (GANs) [8] have proven effective for high-frequency texture synthesis and super-resolution. Previous works [9, 10] have leveraged GANs for sketch-based modeling or style transfer. However, these methods typically treat erosion as a texture in-painting task [11], ignoring the underlying physical laws. A distinct limitation in the current literature is the absence of parametric control; neural terrain models often function as "black boxes," generating plausible topography that cannot be tuned by physical variables such as fluid inertia or sediment capacity.

We propose a **Neural Surrogate Model** designed explicitly to approximate the operators of hydraulic erosion. Rather than replacing physics with hallucinated detail, we train a Conditional GAN (cGAN) [12] to replicate the discrete time-stepping of a Lagrangian simulation. Our primary technical contribution is a novel conditioning mechanism: we spatially broadcast scalar physical parameters and concatenate them with the input topography. This architecture compels the network to learn the specific response of the terrain to varying fluid dynamics properties, bridging the gap between simulation and generation.

We validate this framework against a ground-truth dataset generated via a JIT-compiled particle simulation [6]. Benchmarks reveal a $193\times$ speedup in latency-critical scenarios ($N = 1$) and a $5\times$ throughput advantage in large batches ($N = 64$). Furthermore, the model avoids the mode collapse and artifact accumulation typical of autoregressive generation; recursive inference successfully reproduces the cumulative effects of long-duration erosion (10^6 drops). This stability establishes the method as a robust numerical integrator suitable for real-time geological simulation.

2 Methodology

The proposed framework creates a surrogate model for hydraulic erosion by mapping an initial height-map state to its eroded equilibrium, conditioned on governing physical parameters. The pipeline comprises three stages: procedural generation of priors, ground-truth physics simulation, and the training of a multi-channel cGAN [6, 8, 12].

2.1 Initial Terrain Generation

Generalization across diverse geological features required a dataset of $N = 10,000$ synthetic height maps. We utilized Perlin noise layered via fractional Brownian motion (fBm) to generate continuous, coherent terrain structures. For a grid coordinate (x, y) , the elevation $H(x, y)$ follows [2]:

$$H(x, y) = \sum_{k=0}^{O-1} A \cdot P(f \cdot x, f \cdot y) \quad (1)$$

where P is the noise function, O represents octaves, f is frequency (lacunarity $\lambda \approx 2.0$), and A is amplitude (persistence $\rho \in [0.4, 0.6]$). We applied domain randomization to the scale, offset, and seed of every sample to prevent the network from overfitting to specific frequency artifacts.

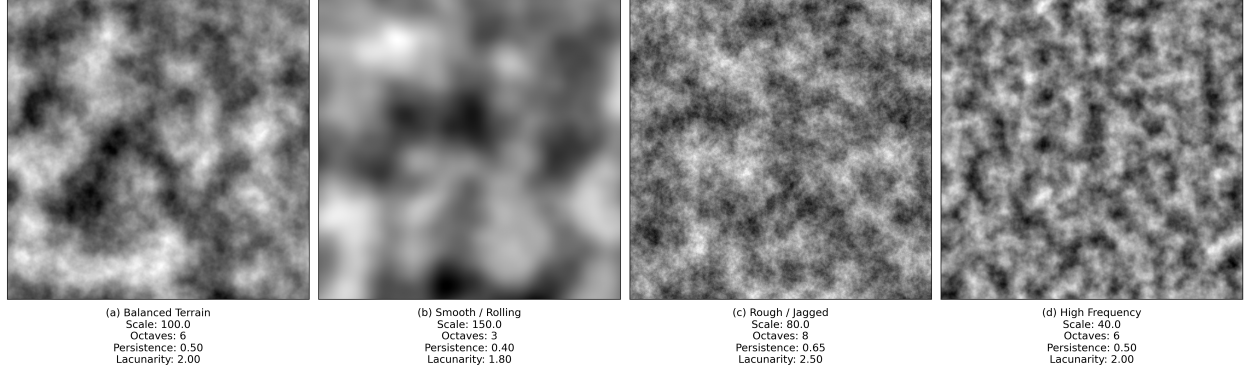


Figure 1: Procedural Priors. Height-maps generated via fractional Brownian motion, illustrating the domain randomization used to train the network on diverse topographical features.

2.2 Physics-Based Ground Truth Simulation

We generated ground truth targets using a Lagrangian particle-based erosion simulation derived from Beyer [6]. To process the large dataset efficiently, we implemented the algorithm in Python using Numba JIT (Just-In-Time) compilation.

2.2.1 Particle Motion

The simulation iterates $N_{drops} = 200,000$ water particles per map. Unlike Eulerian grid-based fluids, particles move freely across height-map coordinates. The trajectory combines the local gradient $\vec{g}$ and previous velocity, blended by an inertia parameter $p_{inertia}$. The update vector $\vec{d}_{new}$ is [6]:

$$\vec{d}_{new} = \vec{d}_{old} \cdot p_{inertia} - \vec{g} \cdot (1 - p_{inertia}) \quad (2)$$

2.2.2 Sediment Transport Capacity

Sediment transport capacity C determines how much soil a drop carries. We model this as a function of local slope magnitude (Δh), velocity (v), and water volume (w), scaled by a global capacity factor $p_{capacity}$ [6]:

$$C = \max(-\Delta h, p_{minSlope}) \cdot v \cdot w \cdot p_{capacity} \quad (3)$$

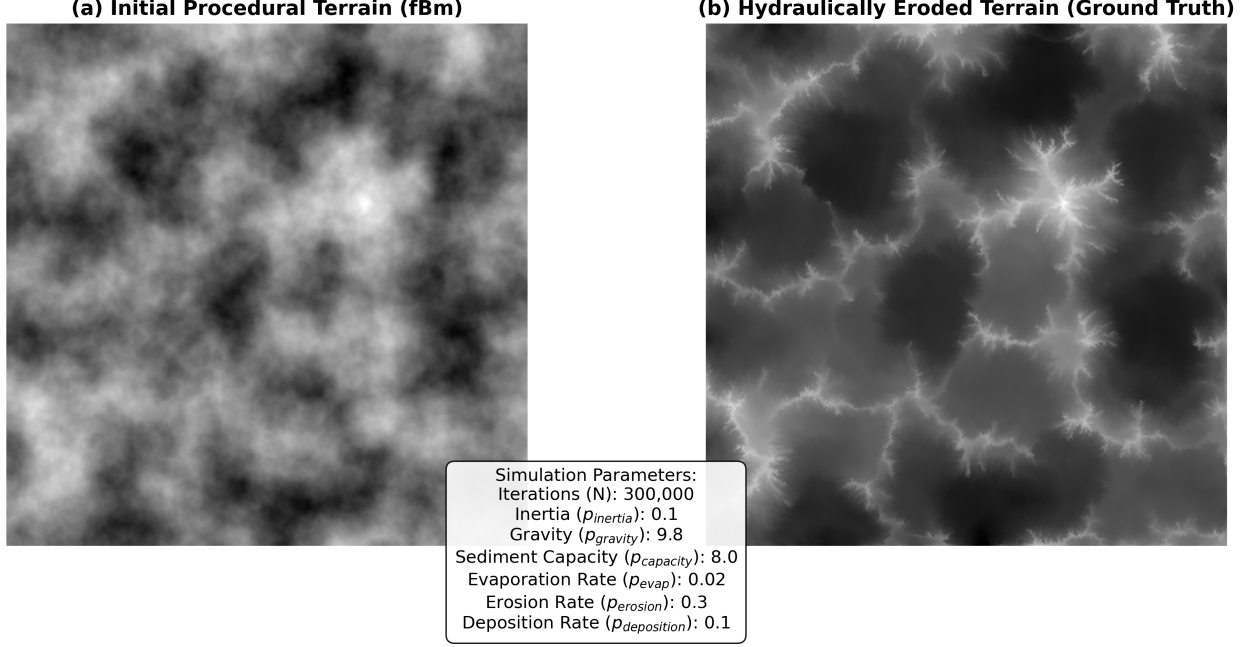


Figure 2: Ground Truth Generation. Comparison of the original noise map (left) and the eroded terrain after the Lagrangian physics simulation (right), showing the formation of dendritic drainage networks.

2.2.3 Mass Transfer

The simulation modifies the height-map based on the difference between the particle’s current sediment load (s) and its capacity (C).

- **Deposition:** If $s > C$, the particle deposits a fraction of the surplus sediment, defined by $p_{deposition}$:

$$\Delta s_{dep} = (s - C) \cdot p_{deposition} \quad (4)$$

- **Erosion:** If $s < C$, the particle dissolves terrain from the current node. We limit erosion to the available height difference to prevent pit formation:

$$\Delta s_{ero} = \min((C - s) \cdot p_{erosion}, -\Delta h) \quad (5)$$

2.2.4 Velocity and Evaporation

Post-transfer, velocity updates based on potential-to-kinetic energy conversion (gravity $g = 9.8$), and water volume decreases via evaporation p_{evap} :

$$v_{new} = \sqrt{v_{old}^2 - \Delta h \cdot g} \quad , \quad w_{new} = w_{old} \cdot (1 - p_{evap}) \quad (6)$$

2.3 Conditional GAN Architecture

We adapted the Pix2Pix framework [12] to support multi-variable parameter conditioning. Figure 3 details the architecture.

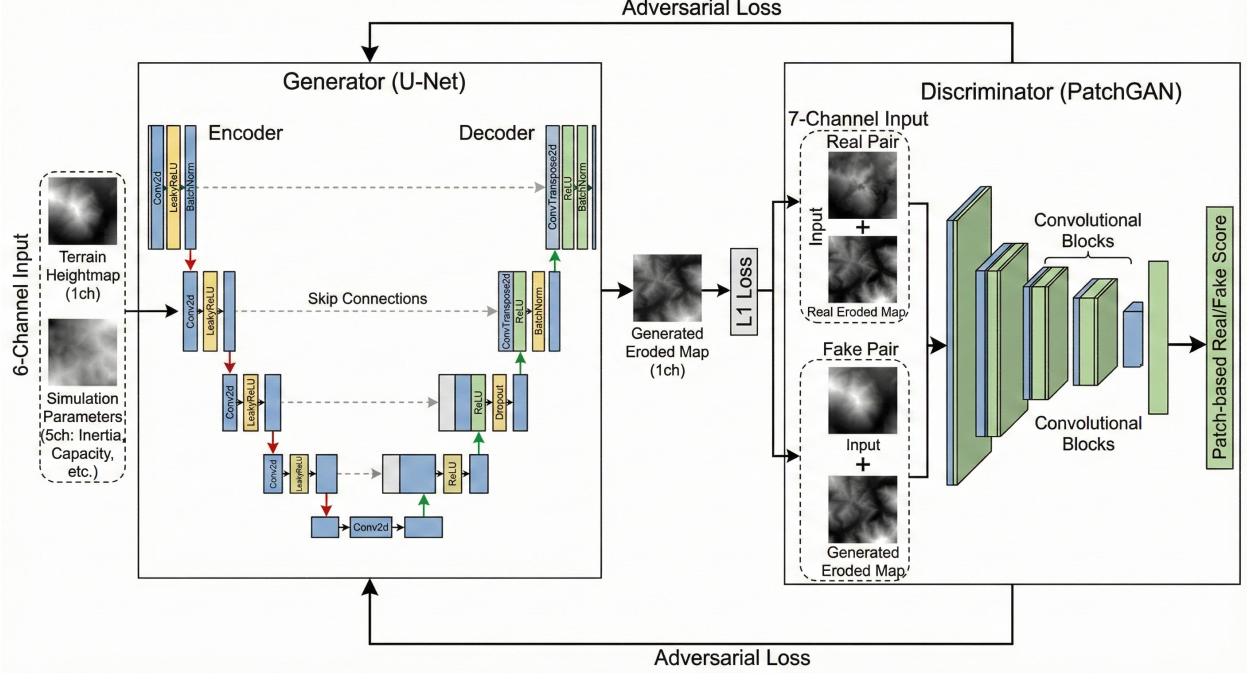


Figure 3: cGAN Architecture. The Generator (U-Net) is conditioned on a 6-channel input tensor containing the heightmap and spatially broadcasted physics parameters. The PatchGAN discriminator evaluates local textural realism.

2.3.1 Input Representation

Standard cGANs condition solely on the input image [8]. To inject physics control, we construct a 6-channel input tensor $\mathbf{X} \in \mathbb{R}^{6 \times 512 \times 512}$.

- **Channel 1:** The uneroded height-map $H_{initial}$.
- **Channels 2-6:** Spatial broadcast of scalar simulation parameters ($p_{inertia}$, $p_{capacity}$, p_{evap} , $p_{erosion}$, $p_{deposition}$).

This spatial expansion allows convolutional filters in the Generator’s initial layers to correlate local terrain features directly with global physical constraints.

2.3.2 Generator and Discriminator

The Generator uses a U-Net architecture with symmetric skip connections [13] to preserve high-frequency structural details (ridges) while synthesizing eroded textures. For the Discriminator, we employ a PatchGAN [12], which evaluates the realism of $N \times N$ local image patches. This enforces high-frequency textural accuracy, ensuring generated ravines are indistinguishable from ground truth.

2.4 Training Strategy

We trained the network using a composite objective function combining adversarial loss ($\mathcal{L}_{cGAN}$) and pixel-wise L1 reconstruction loss ($\mathcal{L}_{L1}$) [8].

$$G^* = \arg \min_G \max_D \mathcal{L}_{cGAN}(G, D) + \lambda \mathcal{L}_{L1}(G) \quad (7)$$

We used a regularization parameter $\lambda = 100$ to prioritize structural accuracy. The model trained for 100 epochs using the Adam optimizer [14] ($lr = 0.0002$, $\beta_1 = 0.5$), while utilizing Automatic Mixed Precision (AMP) to maximize GPU throughput on 512×512 inputs.

3 Results and Discussion

We evaluated the parameter-conditioned cGAN across three critical dimensions: computational scalability, geomorphological fidelity relative to Lagrangian ground truth, and temporal stability during recursive inference. All deep learning inference ran on a consumer-grade GPU (AMP enabled), compared against the Numba-optimized CPU physics baseline.

3.1 Computational Efficiency

Lagrangian particle simulations suffer from linear time complexity, $O(N_{drops})$, where execution time rises proportionally with erosion severity. In contrast, the cGAN operates with constant time complexity, $O(1)$. To quantify this, we benchmarked scalability across batch sizes $N \in \{1, \dots, 64\}$ on 512×512 heightmaps. Figure 4 reveals two performance regimes.

For single-instance inference ($N = 1$), the surrogate model achieves a speedup factor of $193\times$ (10ms vs. 1.93s). This reduces the feedback loop from seconds to sub-frame latency, enabling real-time hydraulic sculpting. As concurrent requests rise ($N \geq 32$), hardware constraints induce serialization, yet the model maintains a speedup factor between $3\times$ and $5\times$. This represents a fundamental decoupling of computational cost from simulation complexity; simulating a light drizzle (10k drops) or a heavy monsoon (1M drops) incurs the exact same neural inference cost.

3.2 Geomorphological Feature Reconstruction

Visual analysis confirms the network has learned the underlying operator of the physical equations rather than a simple style transfer. Figure 5 presents a comparative analysis. The cGAN successfully reconstructs complex dendritic drainage networks that align spatially with the physics-based ground truth. Notably, the model demonstrates sensitivity to broadcasted physical parameters. Lower inertia values ($p_{inertia}$) correctly result in highly tortuous river paths, while higher values produce straighter channels. Similarly, the network respects mass conservation; in regions where slope gradient ∇H approaches zero, the model correctly synthesizes sediment accumulation (p_{dep}), effectively approximating the capacity limit.

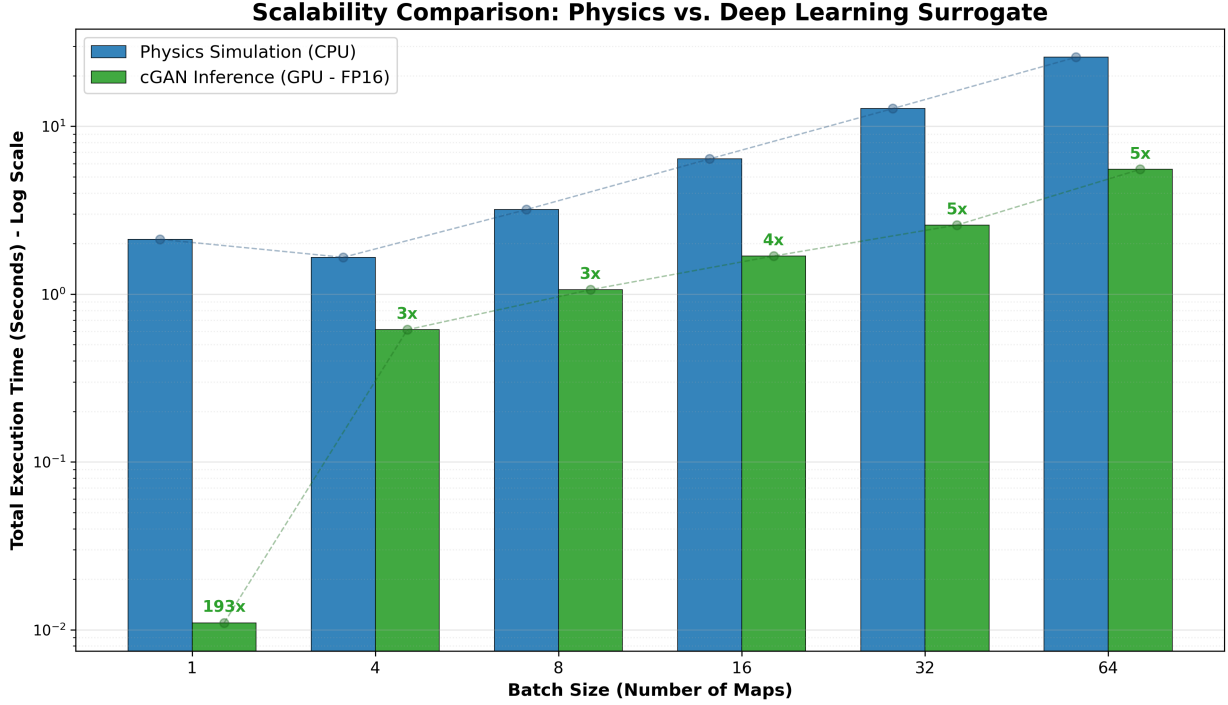


Figure 4: Scalability Analysis. Comparison of execution time between Physics Simulation (CPU) and cGAN (GPU). The log-scale highlights the 193 $\times$ speedup for interactive applications.

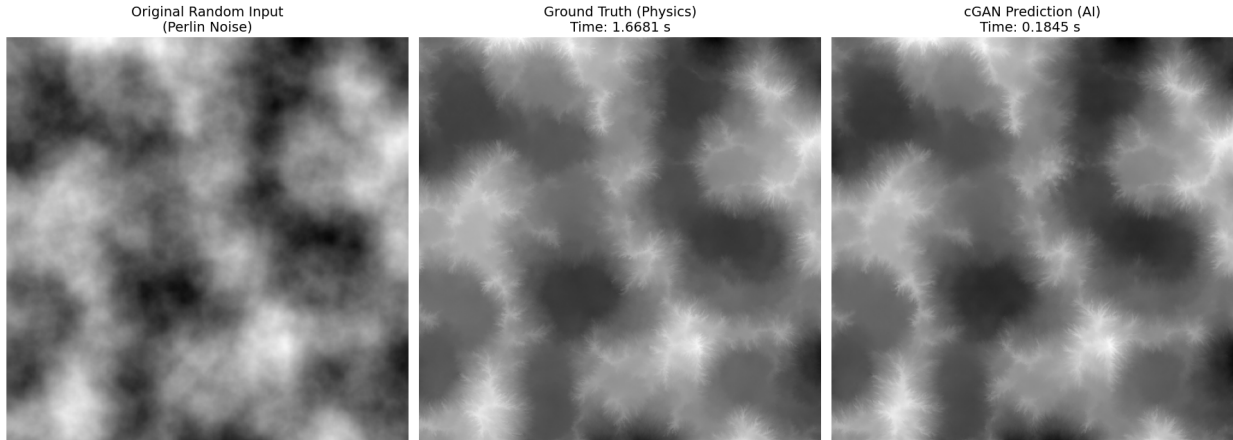


Figure 5: Qualitative Evaluation. Parameter sensitivity and Ground Truth Comparison. The cGAN (right) faithfully reproduces the erosion channels and deposition zones generated by the physics engine (center).

3.3 Recursive Stability

Autoregressive generation frequently suffers from error accumulation (“drift”), where minor artifacts in step t amplify by step $t + 1$. To evaluate robustness, we tasked the model with

a recursive simulation of 1×10^6 droplets, achieved by feeding the cGAN’s output back as input for 5 consecutive iterations.

The results (Figure 6) demonstrate exceptional stability. The Mean Squared Error (MSE) relative to the cumulative physics simulation grew linearly rather than exponentially:

$$\text{MSE}_{t=1} = 1.45 \times 10^{-4} \quad \rightarrow \quad \text{MSE}_{t=5} = 3.67 \times 10^{-4}$$

Error heatmaps indicate deviations are structurally localized to high-frequency ridge lines—a classic discrete sampling challenge—while sediment deposition basins maintain near-perfect fidelity. This confirms the network has converged to a stable numerical operator for the erosion PDE [15], permitting its use as a time-stepped simulator.

4 Conclusion

This work introduces a deep learning framework for the real-time approximation of hydraulic erosion, addressing the computational bottleneck of Lagrangian particle hydrodynamics. By conditioning a U-Net generator on spatially broadcasted physical parameters, we created a differentiable surrogate model that retains user control over fluid dynamics properties while eliminating the iterative cost of particle tracking.

The proposed method achieves a $193\times$ reduction in latency for interactive applications, enabling physics-based terrain editing at interactive framerates. Empirical validation confirms the model’s recursive stability, allowing for the simulation of extensive geological timeframes without significant mode collapse. While currently limited to 2.5D heightfield representations, this parameter-conditioning strategy establishes a precedent for neural surrogate modeling in geomorphology. Future work will investigate the extension of this architecture to volumetric data (3D Voxel grids) and the integration of multi-modal weathering processes, such as thermal and glacial erosion.

References

- [1] B. B. Mandelbrot, *The Fractal Geometry of Nature*. W. H. Freeman and Company, 1982.
- [2] K. Perlin, “An image synthesizer,” *SIGGRAPH Computer Graphics*, vol. 19, no. 3, pp. 287–296, 1985.
- [3] F. K. Musgrave, C. E. Kolb, and R. S. Mace, “The synthesis and rendering of eroded fractal terrain,” in *Proc. SIGGRAPH ’89*, 1989, pp. 41–50.
- [4] N. Chiba, K. Muraoka, and K. Fujita, “An erosion model based on velocity fields for the visual simulation of mountain scenery,” *The Journal of Visualization and Computer Animation*, vol. 9, no. 4, pp. 185–194, 1998.
- [5] B. Beneš, V. Těšínský, J. Hornyš, and S. K. Bhatia, “Hydraulic Erosion,” *Computer Animation and Virtual Worlds*, vol. 17, no. 2, pp. 99–108, 2006.

- [6] H. T. Beyer, “Implementation of a method for hydraulic erosion,” Bachelor’s thesis, Dept. Informatics, Tech. Univ. Munich, Munich, Germany, 2015.
- [7] V. Skorkovská, I. Kolingerová, and B. Beneš, “Hydraulic Erosion Modeling on a Triangular Mesh,” in *Surface Models for Geosciences*, Springer, 2015, pp. 237–247.
- [8] I. Goodfellow *et al.*, “Generative Adversarial Nets,” in *Advances in Neural Information Processing Systems (NIPS)*, vol. 27, 2014, pp. 2672–2680.
- [9] E. Guérin, J. Digne, E. Galin, and A. Peytavie, “Interactive Example-Based Terrain Authoring with Conditional Generative Adversarial Networks,” *ACM Trans. Graph. (TOG)*, vol. 36, no. 6, pp. 228:1–228:13, 2017.
- [10] G. Cordonnier, E. Galin, J. Gain, B. Beneš, and E. Guérin, “Authoring Landscapes by Combining Ecosystem and Terrain Erosion Simulation,” *ACM Trans. Graph. (TOG)*, vol. 36, no. 4, pp. 134:1–134:12, 2017.
- [11] P. Henzler, N. J. Mitra, and T. Ritschel, “Escaping Plato’s Cave: 3D Shape From Adversarial Rendering,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2019, pp. 9984–9993.
- [12] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros, “Image-to-Image Translation with Conditional Adversarial Networks,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 1125–1134.
- [13] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional Networks for Biomedical Image Segmentation,” in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2015, pp. 234–241.
- [14] D. P. Kingma and J. Ba, “Adam: A Method for Stochastic Optimization,” in *Int. Conf. Learn. Represent. (ICLR)*, 2015.
- [15] Z. Li *et al.*, “Fourier Neural Operator for Parametric Partial Differential Equations,” in *Int. Conf. Learn. Represent. (ICLR)*, 2021.

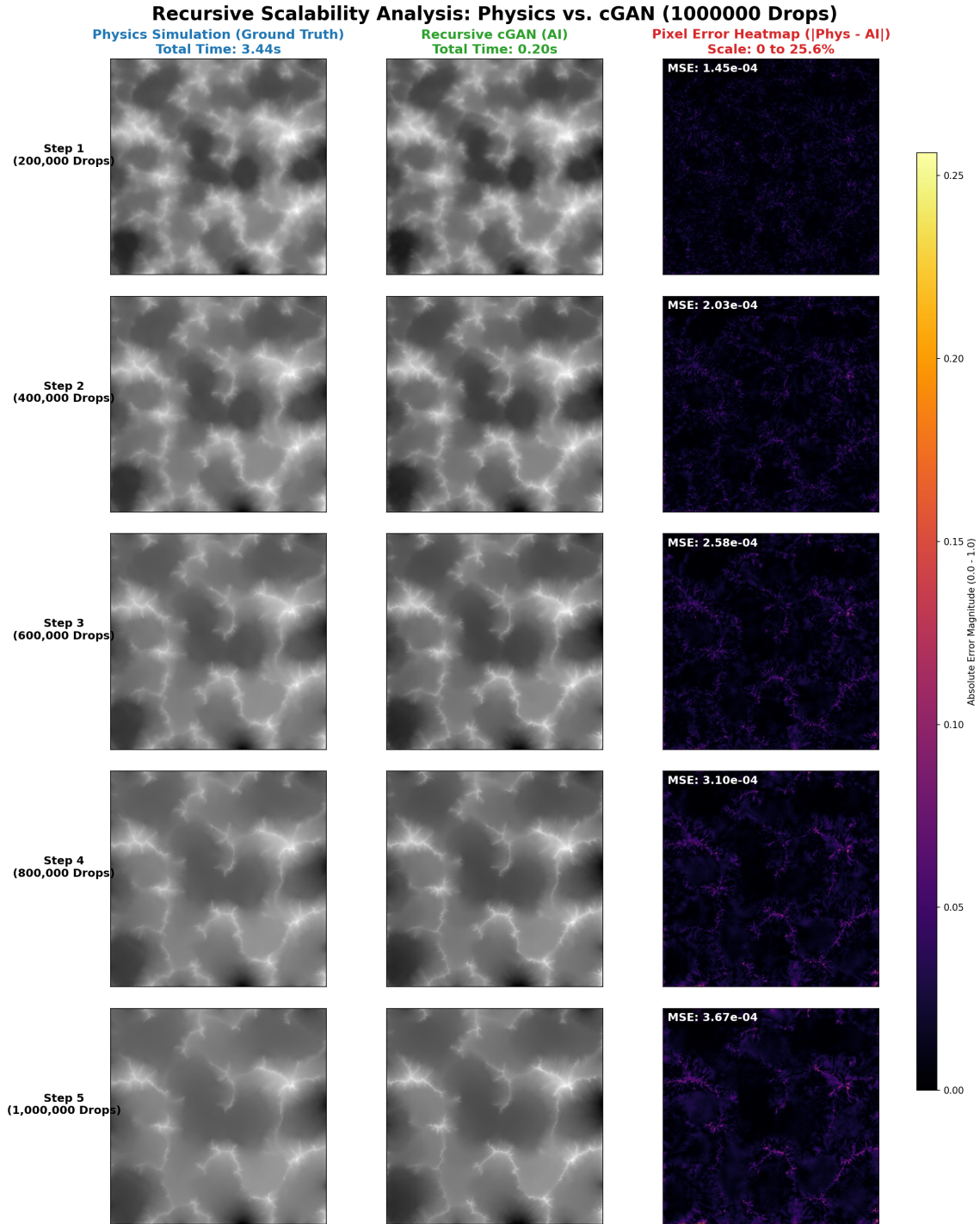


Figure 6: Recursive Inference and Error Propagation. A temporal stability test simulating 1 million drops. The error heatmaps (right) show that the MSE remains low even after 5 recursive passes.